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Presentation Video Link:

<https://drive.google.com/drive/folders/1npdFA0IE-WBRfEMvUUVxLmp7KCEqHUG9?usp=sharing>

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PROBLEM STATEMENT

Every student in traditional educational systems adheres to the same learning structure, regardless of their learning style, abilities, or areas of weakness. This "one-size-fits-all" approach creates serious problems in the classroom, where slower learners find it difficult to keep up and rapid learners may lose interest. Many students score poorly on exams and lack enthusiasm because they are unaware of their specific knowledge deficiencies until it is too late.

Additionally, teachers find it challenging to pinpoint students' areas of weakness, particularly when overseeing large classrooms. Because manual tracking of student development mostly relies on subjective observation rather than data-driven analysis, it is time-consuming and frequently erroneous.

Furthermore, most existing e-learning platforms place a higher priority on material distribution than on adaptive learning. They don't provide customised learning paths that adapt to each student's performance, and they don't track students' advancement in real time.

An AI-powered system that can automatically evaluate student performance, identify knowledge gaps, and suggest customised learning paths is therefore required. By offering real-time insights, such a system would benefit instructors and students alike, guaranteeing a more effective, interesting, and customised learning environment.

AI SOLUTION

To solve the limitations of traditional learning systems, in which the students take the same path of learning regardless of knowing about their strengths or weaknesses, we proposed the AI adaptive learning system. In this way, the system allows the students to learn more effectively by assessing their performance on tests, assignments, and learning behaviors such as time spent on the lectures and number of attempts. The system applies an AI algorithm to obtain knowledge gaps and indicate specific areas of improvement for the students.

Through this analysis, the system can automatically come up with a personalized learning path. It offers lessons, videos, tutorials, and further resources based on the student's present level of knowledge. The learning route is totally adaptive where it will update in real-time as students progress so that it keeps the focus on the most crucial topics.

The system consists of activity tracking dashboards, evaluation metrics, and a graphical knowledge map that shows the students' proficiency levels among others. With this, the instructors can have a clear view not only of the single learners but of the whole class which enables them to monitor the learners' progress and spot the common weaknesses. This approach can offer a customized, engaging, and research-based learning process that takes the student and the lecturer into account through the merging of insightful evaluation and flexible lesson planning.

GOAL OF AI SOLUTION

The main goal of this project is to improve learning efficiency and understanding by creating a system of education that is data-driven, flexible, and customized for each student. Particularly, the system aims to:

- 1. Identify and overcome knowledge gaps early.**

The system provides continuous performance tracking and intelligent data analysis. By analyzing students' quiz and test marks, the AI will analyze which topic the students are lacking and instantly calculate the target marks they need to achieve an A.

- 2. Personalize learning modules.**

After analyzing the topic the students are lacking, AI will suggest some modules or materials for the student to improve on. The AI will keep track of students' learning journey and by the end of the modules, the AI will generate a trial exam based on past year questions provided by the lecturers. This process helps students to identify their current understanding and overall performance. Therefore, it leads to more consistent and long-term understanding of their courses.

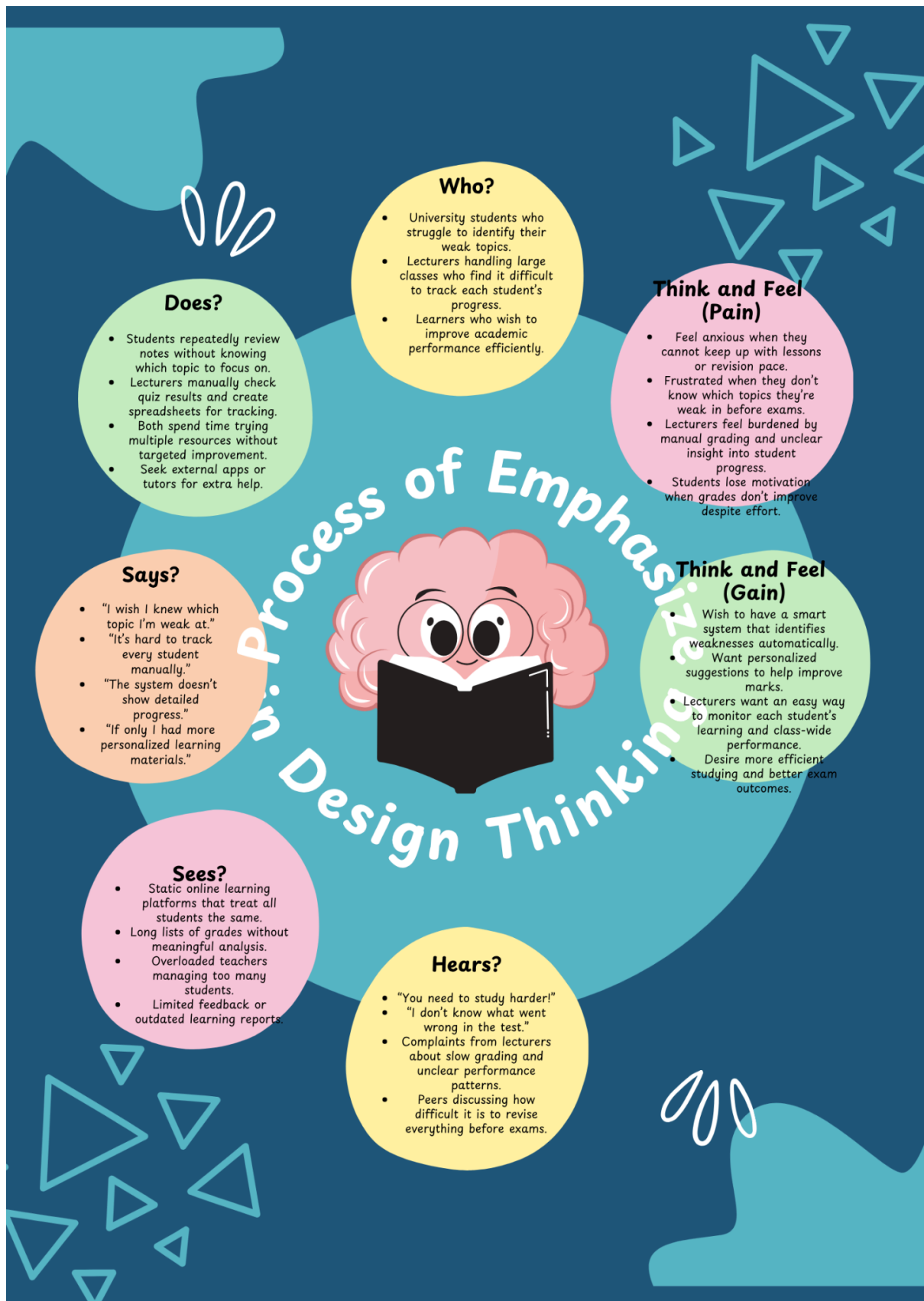
- 3. Assist lecturers in tracking each student's progress effectively.**

AI will help lecturers to track each student's progress by having an access to monitor what their students' weaknesses are, including weak topics and their effort to improve on what they are lacking. At the end of the trial exam, lecturers could identify whether their student is prepared or not for the actual final examination.

- 4. Reduce the gap between students who have difficulties in understanding and those who are advanced.**

With the help of AI, students who are struggling could work harder in their studies as this AI already identifies their weaknesses and provides some materials for them to work on so that they could get an A in their courses and be on par with the advanced students.

PROCESS OF EMPATHIZE IN DESIGN THINKING



PROCESS OF DEFINE IN DESIGN THINKING

After developing empathy for instructors and students, the team determined the fundamental problems that must be solved and precisely described the issue to direct the creation of a successful AI solution.

User Persona

- i) Student Persona: Students frequently struggle to identify their areas of weakness. To increase their comprehension, they require a system that can evaluate their performance, pinpoint their areas of weakness and provide them with tailored learning resources.
- ii) Lecturer Persona: It might be difficult for lecturers to keep track of each student's progress, particularly in large classrooms. In order to concentrate their instruction on subjects where students struggle the most, they want an AI tool that summarises pupils' strengths and weaknesses.

Problem Statement (Defined)

“Lecturers need a way to effectively monitor student progress and performance trends while students need an intelligent learning system that can automatically identify their knowledge gaps and provide a personalised learning path.”

Key Insights

- Students desire continuous feedback and adaptive learning rather than fixed study materials.
- Instructors are looking for an automated way to find the class's common weaknesses.
- An intuitive interface that shows advancement and learning gains is valued by both sides.

Design Requirements

- 1) AI algorithm for identifying knowledge gaps in test and quiz data.
- 2) Personalized learning path generator that adapts based on progress.
- 3) Lecturers can monitor students' performance analytics on this dashboard.
- 4) Interactive visualisation of progress like charts or knowledge maps.
- 5) Real-time updates that adjust educational resources based on improvements.

Knowledge Representations

1. Student Knowledge Profile

This knowledge representation stores every student's proficiency in every topic. It helps the system to recognise weak and strong areas, and detect knowledge gaps. This KR is important for personalized learning, as it produces the core information for the AI to decide what kind of learning resources needed for the students.

i) *IF x is in the student database, THEN x is a student*

ii) *IF y is listed in the course structure, THEN y is a topic.*

iii) *IF $\text{Proficiency}(x,y,l)$ is true, THEN student x has proficiency score of l in topic y .*

iv) *IF there exist a proficiency level l such that $\text{Proficiency}(x,y,l)$ is true AND l is less than 40, THEN x is weak in topic y*

v) *IF there exist a proficiency level l such that $\text{proficiency}(x,y,l)$ is true AND is more than 80, THEN x is strong in topic y .*

vi) *IF a student is not weak AND not strong in a topic, AND their proficiency falls between 40 and 80, THEN x is moderate in topic y .*

Explanation:

The student knowledge profile evaluates each student's understanding of every topic by assigning a proficiency score. If the score is below than 40, the student is marked as weak in that topic, showing that this student needs additional guidance. When the score meets or exceeds 80, the student is categorized as strong, indicating that the student is ready for advanced materials. And the student is marked as moderate in that topic if the score falls between 40 and 80. By identifying the proficiency levels for every student, the system can accurately analyze learning gaps and prepare a suitable recommendation such as practice, modules, or learning videos.

Predicate Logic/First Order Logic:

i) $\forall x (\text{InStudentDatabase}(x) \rightarrow \text{Student}(x))$

ii) $\forall y (\text{InCourseStructure}(y) \rightarrow \text{Topic}(y))$

iii) $\forall x \forall y \forall l (\text{Proficiency}(x,y,l) \rightarrow \text{Student}(x) \wedge \text{Topic}(y))$

iv) $\forall x \forall y (\text{Weak}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge l < 40))$

v) $\forall x \forall y (\text{Strong}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge l \geq 80))$

vi) $\forall x \forall y (\text{Moderate}(x,y) \leftrightarrow \exists l (\text{Proficiency}(x,y,l) \wedge 50 \leq l < 80))$

For every x that exists in the student database, then x is a student.

For every y that exists in course structure, then y is a topic.

For every student x , topic y , and every level l , if x has proficiency level l for topic y , then x must be a student and y must be a topic.

For every student x and every topic y , x is weak in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is less than 50.

For every student x and every topic y , x is strong in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is more than 80.

For every student x and every topic y , x is moderate in topic y , if and only if there exists a proficiency level l such that x has proficiency level l in y and l is greater or equal to 50, but less than 80.

2. Personalize Learning Path

It makes predictions about the learning resources based on the student knowledge profile that considers their skills, the areas where they are lacking, and their grasp of the basic topics so that their development is at the right speed and the right learning techniques are used.

i) *IF a student is weak in a topic THEN recommend a remedial module.*

ii) *IF a student is moderate in a topic THEN recommend a practice module.*

iii) *IF a student is strong in a topic AND the student is weak in the next topic THEN recommend the next topic module.*

Explanation:

If a student has trouble with a subject, the system will first evaluate the student's knowledge and skills in related subjects before recommending the most appropriate course material. In the situation where the students have a weak understanding on the topic, this approach will offer a remedial module designed to reinforce their basic concepts. If the students have moderate proficiency, the AI system will offer them a practice module that will aid them to comprehend better and gain skills. If the student shows a strong understanding on the topic, the AI system will offer the next topic module. This approach helps each student get a personalized learning way that leads them through content as their needs and speed by smartly choosing learning materials depending on the student's ability.

Predicate Logic/First Order Logic:

i) $\forall x \forall y ((\text{weak}(x,y)) \rightarrow \text{recommend}(x, \text{remedial_module}(y)))$

ii) $\forall x \forall y ((\text{moderate}(x,y)) \rightarrow \text{recommend}(x, \text{practice_module}(y)))$

iii) $\forall x \forall y \forall z ((\text{strong}(x,y) \wedge \text{subtopic}(y,z) \wedge \text{weak}(x,z)) \rightarrow \text{recommend}(x, \text{next_topic_module}(z)))$

For every student x and every topic y , if the student x is weak in topic y then a remedial module for topic y is recommended to the student.

For every student x and every topic y , if the student x is moderate in topic y then a practice module for topic y is recommended to the student.

For every student x , every main topic y , and every related next topic z , if the student x is strong in the main topic y and weak in the next topic z , then the next topic module for y is recommended to the student.

3. Real-time Engagement Monitor

This knowledge representation tracks student engagement and attention levels while they are learning to identify the times students are losing their focus, getting confused, or not engaging well with the material so the system can intercede at the right moment to keep them on track.

i) IF a student's response time increases significantly AND answer accuracy decreases THEN alert for possible confusion or fatigue.

ii) IF a student repeatedly skips questions or leaves sections incomplete AND time spent per question is very short THEN detect disengagement and recommend motivational break.

iii) IF a student shows consistent engagement (regular progress, steady response time) AND maintains good accuracy THEN continue current learning path without interruption.

iv) IF a student's engagement suddenly drops after specific content type (video/text/exercise) THEN flag that content format as less effective for that student.

v) IF a student attempts same question multiple times AND time spent increases each attempt THEN identify struggling point and offer hint or alternative explanation.

Explanation:

The real time engagement monitor keeps track of how students are actually interacting with their given learning materials while they are working on them. This is not just about getting the answers right or wrong, the system picks up on behavioral patterns showing someone is struggling. When someone takes longer time to answer questions and at the same time the accuracy of the answers drops, that means they are either confused or losing focus. The system will catch this early and alerts changes needed before the student gets frustrated. When students start rushing through the material, skipping parts, or barely spending time before moving on to the next part, that is a clear signal that they are mentally checked out. The

system will detect this disengagement trend and suggest stepping away for a bit and taking a break.

The system also notices types of content work for each student. If engagement consistently drops right after watching videos or reading text sections, the information and data will tell the system those formats are not effective for that specific student. It can then adjust recommendations to show them content in formats they respond to way better. The system will recognize the struggles when the student keeps attempting the same question multiple times with a helpful hint or offer explanations.

Predicate Logic/First Order Logic:

- i) $\forall s \quad \forall t \quad ((\text{ResponseTimeIncrease}(s,t) \quad \wedge \quad \text{AccuracyDecrease}(s,t)) \rightarrow \text{AlertConfusionOrFatigue}(s,t))$
- ii) $\forall s \quad \forall t \quad ((\text{FrequentSkipping}(s,t) \quad \wedge \quad \text{VeryShortTimePerQuestion}(s,t)) \rightarrow (\text{DetectDisengagement}(s,t) \wedge \text{RecommendBreak}(s)))$
- iii) $\forall s \quad \forall t \quad ((\text{ConsistentEngagement}(s,t) \wedge \text{GoodAccuracy}(s,t)) \rightarrow \text{ContinueCurrentPath}(s,t))$
- iv) $\forall s \quad \forall c \quad \forall t \quad ((\text{EngagementDropAfterContent}(s,c,t)) \rightarrow \text{FlagIneffectiveFormat}(s,c))$
- v) $\forall s \quad \forall q \quad ((\text{MultipleAttempts}(s,q) \quad \wedge \quad \text{IncreasingTimePerAttempt}(s,q)) \rightarrow (\text{IdentifyStrugglingPoint}(s,q) \wedge \text{OfferHint}(s,q)))$
- vi) $\text{ResponseTimeIncrease}(s,t) \leftrightarrow \exists r1 \quad \exists r2 \quad (\text{PreviousResponseTime}(s,t,r1) \quad \wedge \quad \text{CurrentResponseTime}(s,t,r2) \wedge r2 > 1.5 \times r1)$
- vii) $\text{ConsistentEngagement}(s,t) \leftrightarrow (\text{RegularProgress}(s,t) \wedge \text{SteadyResponseTime}(s,t) \wedge \neg \text{FrequentSkipping}(s,t))$

For every student s and topic t , if the student's response time increases significantly and answer accuracy decreases simultaneously, then the system alerts for possible confusion or fatigue.

For every student s and topic t , if the student frequently skips questions and spends very little time on each question, then disengagement is detected and a motivational break is recommended.

For every student s and topic t , if the student shows consistent engagement patterns and maintains good accuracy, then the current learning path continues without interruption.

For every student s , content type c , and topic t , if engagement suddenly drops after encountering specific content type, then that content format is flagged as less effective for that student.

Response time is considered significantly increased if the current response time is more than 1.5 times the previous average response time.

Consistent engagement exists if and only if the student demonstrates regular progress, maintains steady response time, and does not frequently skip content.

4. Adaptive Difficulty Progression Tracker

This knowledge representation monitors and adjusts the difficulty level of learning materials and assessments in real-time based on each student's performance path, ensuring struggling students receive appropriately suitable content while advanced students are continuously challenged with harder materials to prevent boredom and maintain their engagement towards learning. This KR specifically focuses on adjusting the challenge level within the learning materials, working in tandem with Personalized Learning Path KR which determines what type of content the student needs.

- i) IF a student consistently scores below 40% on assessments for a topic AND shows no improvement over 3 attempts THEN reduce difficulty level and provide simplified content with guided examples.*
- ii) IF a student consistently scores above 85% on current difficulty level AND completes tasks quickly THEN increase difficulty level and introduce advanced challenge materials.*
- iii) IF a struggling student reaches 65% in previously weak topic THEN unlock intermediate-level content to bridge gap toward advanced materials.*

Explanation:

The adaptive difficulty progression tracker closes the performance quality gap between struggling and advanced students by making sure everyone works at the correct specified challenge level. Unlike the Personalized Learning Path which decides whether students need remedial, or practice, this KR determines how challenging those materials should be based on the ongoing performance trends. When students keep scoring poorly after multiple attempts, the system recognizes they need simpler content with step-by-step examples rather than repeating the same difficult material. This prevents them from falling further behind and builds their confidence with achievable goals.

For students with high scores and quick completion, the system raises difficulty and introduces complex problems to keep them engaged. This step prevents advanced students from getting bored and disengaged. The tracker also monitors students gradually improving from weak to

moderate to high performance. It carefully increases difficulty in small steps aligned with their improvement scale, helping them learn without sudden risky jumps. Once struggling students reach moderate proficiency that is needed, the system unlocks intermediate content bridging them toward advanced materials. Even for advanced students, if performance drops and time increases significantly on harder content, the system backs off to find that optimal challenge zone where work is engaging but not frustrating. This dynamic difficulty adjustment ensures that once learning path is determined, the materials presented at each stage are optimally calibrated to each student's current capability level.

Predicate Logic/First Order Logic:

- i) $\forall s \forall t ((\text{ConsistentlyLowScore}(s,t) \wedge \text{NoImprovement}(s,t,3)) \rightarrow (\text{ReduceDifficulty}(s,t) \wedge \text{ProvideSimplifiedContent}(s,t)))$
- ii) $\forall s \forall t ((\text{ConsistentlyHighScore}(s,t) \wedge \text{QuickCompletion}(s,t)) \rightarrow (\text{IncreaseDifficulty}(s,t) \wedge \text{IntroduceAdvancedMaterials}(s,t)))$
- iii) $\forall s \forall t ((\text{GradualImprovement}(s,t) \wedge \text{ConsistentEffort}(s,t)) \rightarrow \text{ProgressivelyIncreaseDifficulty}(s,t))$
- iv) $\forall s \forall t ((\text{PreviouslyWeak}(s,t) \wedge \text{CurrentlyModerate}(s,t)) \rightarrow \text{UnlockIntermediateContent}(s,t))$
- v) $\forall s \forall t ((\text{AdvancedStudent}(s) \wedge \text{DecliningPerformance}(s,t) \wedge \text{TimeIncreaseSignificant}(s,t)) \rightarrow \text{ReduceToOptimalLevel}(s,t))$
- vi) $\text{ConsistentlyLowScore}(s,t) \leftrightarrow \forall a (\text{RecentAssessment}(a,t) \wedge \text{TakenBy}(s,a) \rightarrow \text{Score}(s,a) < 50)$
- vii) $\text{ConsistentlyHighScore}(s,t) \leftrightarrow \forall a (\text{RecentAssessment}(a,t) \wedge \text{TakenBy}(s,a) \rightarrow \text{Score}(s,a) \geq 85)$
- viii) $\text{GradualImprovement}(s,t) \leftrightarrow \exists a1 \exists a2 \exists a3 (\text{Assessment}(a1,t) \wedge \text{Assessment}(a2,t) \wedge \text{Assessment}(a3,t) \wedge \text{Score}(s,a1) < \text{Score}(s,a2) < \text{Score}(s,a3))$

For every student s and topic t , if the student consistently scores below 50% and shows no improvement over 3 attempts, then difficulty is reduced and simplified content with guided examples is provided.

For every student s and topic t , if the student consistently scores above 85% and completes tasks quickly, then difficulty is increased and advanced challenge materials are introduced.

For every student s and topic t , if the student shows gradual improvement and maintains consistent effort, then difficulty is progressively increased while maintaining appropriate pace.

For every student s and topic t , if the student was previously weak but currently achieves moderate proficiency, then intermediate-level content is unlocked to bridge toward advanced materials.

A student shows consistently low scores if and only if all recent assessments for that topic result in scores below 50%.

A student shows consistently high scores if and only if all recent assessments for that topic result in scores of 85% or higher.

Gradual improvement exists if and only if there exist three sequential assessments where each subsequent score is higher than the previous one.

5. Instructor and Class Analytics

Determine Weak Subjects and Assist Lecturers in Making Decisions

i) IF average_class_proficiency < threshold OR topic_mastery_low THEN alert_topic = on

ii) IF number_of_weak_students $\geq$ weak_limit THEN notify_lecturer = on

iii) IF topic_rank $\leq$ 3 THEN top_weak_topic = true

iv) IF proficiency_drop_is_large THEN decline_alert = on

Explanation:

Instead of concentrating solely on individual students, the Instructor and Class Analytics knowledge representation models the performance of the entire class. It creates class-level indicators by combining each student's competency, assessment results and learning activity log. These indicators assist instructors in determining which subjects the majority of students find difficult and how the overall performance of the class corresponds to desired learning standards. The method can identify trends like widespread misconceptions or persistent poor scores in particular subjects by computing the average proficiency for each topic and keeping track of the number of students who exhibit weaknesses.

The system automatically sends out signals or notice to the instructor when too many students show poor performance or when the average proficiency for a topic drops below a certain threshold. These warnings indicate that the subject might need to be addressed right away such as by reteaching, adding more exercises, changing the lesson's pace or providing extra resources. Additionally, the system records performance trends over time and ranks topics from strongest to weakest, allowing instructors to discover the most problematic subjects and identify sharp drops in student comprehension which is going to guarantee proactive instructional modifications and better data driven choices to serve the entire class.

Predicate Logic / FOL

- i) $\forall c \forall t (Class(c) \wedge AvgProficiency(c, t, avg) \wedge (avg < ClassThreshold \vee TopicWeak(c, t)) \rightarrow TopicAlert(c, t))$
- ii) $\forall c \forall t (WeakCount(c, t, n) \wedge n \geq WeakLimit \rightarrow NotifyLecturer(c, t))$
- iii) $\forall c \forall t (WeakerTopicRank(c, t, r) \wedge r \leq 3 \rightarrow TopWeakTopic(c, t))$
- iv) $\forall c \forall t \forall t1 \forall t2 (AvgProficiencyAt(c, t, t1, a1) \wedge AvgProficiencyAt(c, t, t2, a2) \wedge DroppedSignificantly(a1, a2) \rightarrow RapidDec)$

For each class (c) and topic (t), if that topic's average proficiency falls below the class threshold ($avg < ClassThreshold$) or if ($TopicWeak(c, t)$) indicates that the topic is weak for the class, then the instructor is subsequently prompted to take action by the system's activation of $TopicAlert(c, t)$.

For every class (c), if the number of weak students in each class(c) and topic(t) exceeds or equals the weakness limit, then the instructor will be notified.

For each class(c) and the topic(t), the topic is regarded as one of the top weak topics if its weakness rating is 3 or less.

If proficiency declines considerably for each class(c), topic(t) and time periods(t1 and t2), the system initiates a rapid decline notice.

Knowledge Representation Involved to Achieve Goals

Knowledge Representation (KR) is essential to smart education because it allows AI systems to comprehend, interpret and apply data about students, learning materials and class performance. KR enables systems to offer individualised learning experiences, efficiently track progress and adjust to individual needs by explicitly expressing student's knowledge, learning paths, assessment results and scheduling limitations.

The AI system seeks to narrow the performance gap between struggling and advanced students, discover and close knowledge gaps early, personalise learning modules and let instructors track students' progress. Every KR component, from Student Knowledge Profile to Instructor and Class Analytics is made to accomplish these goals by logically and effectively organising pertinent data for decision making.

1) Student Knowledge Profile KR → Achieving Goal: Identify and Overcome Knowledge Gaps Early

Every student's competency in every subject, including their areas of strength and weakness as well as how their performance changes overtime are stored in the Student Knowledge Profile KR. The method precisely detects knowledge gaps by using structured predicates like weak, moderate and strong to characterise student abilities. This makes it possible to identify problems early on and offer prompt solutions, guaranteeing that students fill in any gaps before they fall behind and promoting long-term comprehension of the curriculum.

2) Personalized Learning Path KR → Achieving Goal: Personalize Learning Modules

By using the student knowledge profile, prerequisite relationships between topics and rules that determine whether a student needs remedial, practice or next-topic modules. The Personalized Learning Path KR finds the best learning resources for each student by automatically creating highly customised learning paths based on the demands and learning pace of each student by expressing subject dependencies, subtopic structures and logic-based conditions. This method guarantees flexible and focused learning opportunities, assisting students in strengthening their areas of weakness while successfully completing the course materials.

3) Real-time Engagement Monitor KR → Achieving Goal: Identify and Overcome Knowledge Gaps Early

This KR monitors students' attention and engagement during the learning process, spotting instances in when they get disinterested in the subject matter, lose focus, or become confused. The technology identifies problems in real time by tracking behavioural trends, such as longer reaction times with worse accuracy, skipped questions, repeated attempts, or abrupt declines in engagement following particular content categories. When such tendencies are detected, the AI can step in with suggestions, other explanations, motivational pauses, or content format modifications that better fit the student's learning preferences. This proactive strategy guarantees that difficulties are promptly resolved, keeping students from falling behind and promoting ongoing, productive learning.

4) Adaptive Difficulty Progression Tracker KR → Achieving Goal: Reduce the Gap Between Struggling and Advanced Students

In order to fit each student's current competency level, this KR dynamically modifies the difficulty of learning materials and exams based on real-time performance monitoring. While advanced students who score highly and finish tasks quickly are given more difficult materials to keep them engaged, struggling students who consistently score low or show no improvement receive simplified content with guided examples to keep them from falling further behind. The approach bridges the gap towards advanced topics by gradually increasing difficulty for pupils who consistently improve and unlocking intermediate content for those who are progressing from weak to moderate proficiency. This KR guarantees that all students advance at a suitable rate by consistently adjusting the degree of difficulty to each student's performance trajectory. This promotes fair learning outcomes and lessens gaps between struggling and advanced students.

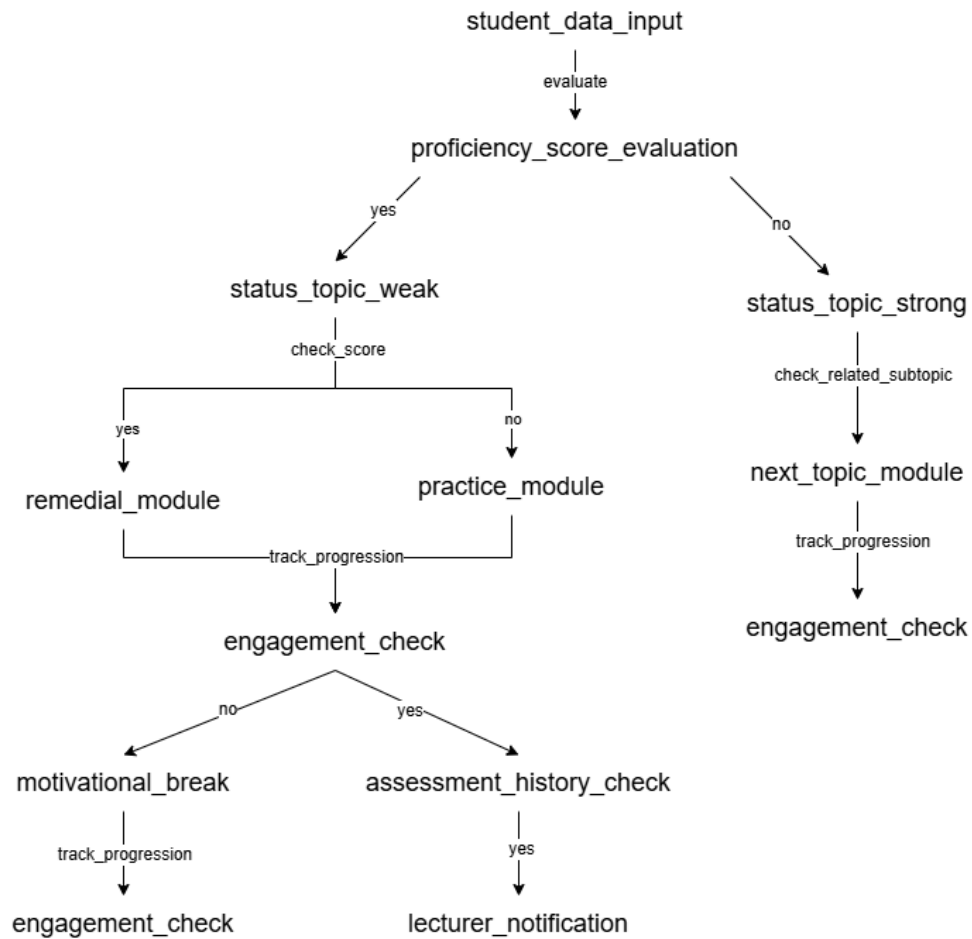
5) Instructor & Class Analytics KR → Achieving Goal: Assist Lecturers in Tracking Each Student's Progress Effectively

Individual student performance is combined into class-level indicators via Instructor and Class Analytics KR. These indicators include average competency, weak topic rankings, broad difficulty alerts and abrupt performance reductions. Instructors or lecturers can rapidly determine which topics the class struggles with, which students require extra attention and which teaching method works best by using ideas like class averages, weak counts, topic alerts and trends over time. Consequently, the instructors are able to effectively track students' progress and make well informed decisions that direct learning, eventually promoting both student achievement and overall class performance.

Topic	AI Analysis (The "Why")	Recommended Action (The "How")
Heuristic Search	68% of students failed the "Admissibility" sub-module.	Deploy Adaptive Quiz: Re-evaluate the concept of with a visual trial exam.
Intelligent Agents	Proficiency dropped significantly this week.	Reteach Session: Release "Agent Environments" video tutorial to 12 weak students.
Search Algorithms	Topic Mastery is low despite high student effort	Resource Push: Add "Dijkstra vs. BFS" practice set to personalized learning paths.

Problem Formulation by State Space Graph

1.0 Smart Education Assistance



1.1 Explanation States and Actions

States	1. student_data_input • The instructor uploads or selects class-related data such as attendance, grades, or assessments. 2. proficiency_score_evaluation
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	• The system processes the class data to evaluate class performance levels.
	3. status_topic_weak • Indicates that the class or a portion of the class is performing below the required threshold.
	4. status_topic_strong • Indicates that the class is performing above the expected standard.
	5. remedial_module • If the students lack prerequisites, the system provides teaching suggestions, remedial strategies, or targeted improvements.
	6. practice_module • If they have the prerequisites, The system provides extra practice to reinforce the current topic.
	7. next_topic_module • The system will provide new and more difficult contents for the students that have mastered the current material.
	8. engagement_check • The system evaluates student engagement (attendance, participation, LMS activity, etc.).
	9. motivational_break • If engagement is low, the system triggers a “break” like a game or video to prevent burnout.

	10. Assessment_history_check • If engagement is high, the system looks at the long-term trend. 11. lecturer_notification • Final state where the system sends analytics results, alerts, or recommendations to the instructor.
Actions	1. analyze • Triggers the system to process and analyse class performance data. 2. yes • A condition is met (e.g., low performance detected, risk identified). 3. no • A condition is not met (e.g., performance above threshold, no risks found). 4. detect_issues • The system examines data to find low-performing areas or problematic topics. 5. recommend • Generates improvement suggestions or interventions for the instructor. 6. alert

	• Sends warnings or flags regarding at-risk or problematic class segments.
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1.2 Problem Formulation and Explanation

Critical	Details	Explanation
Initial State	student_data_input	The initial state of student_data_input implies the system's focus on pulling all the student records and the course structures to begin the evaluation process.
Action	Let $S(x)$ = Set of action-state pairs $S(\text{student_data_input}) = \{\text{evaluate: proficiency_score_evaluation}\}$ $S(\text{proficiency_score_evaluation}) = \{\text{yes:status_topic_weak, no: status_topic_strong}\}$ $S(\text{status_topic_weak}) = \{\text{check_score: (yes: practice_module, no: remedial_module)}\}$ $S(\text{remedial_module}) = \{\text{track_progression: engagement_check}\}$ $S(\text{practice_module}) = \{\text{track_progression: engagement_check}\}$	The process begins with the algorithm that accesses the student information to assess and categorize them according to their present mastery skills. After that, the system starts checks for a low proficiency student by activating check_prereq. The practice_module will be given to the students that have gone through the necessary prerequisites successfully and have mastered them, whereas the non-masters will continue to use the remedial_module. A student who has a solid grasp of the subject matter will be moved to the next_topic_module. The system will then enter a real-time monitoring loop via the engagement_check. If the engagement is maintained, the system will move to evaluate long-term performance through assessment_history_check. If the system

	S(status_topic_strong) = {check_related_subtopic: next_topic_module} S(next_topic_module) = {track_progression: engagement_check} S(engagement_check) = {yes: assessment_history_check, no: motivational_break} S(motivational_break) = {track_progression: engagement_check} S(assessment_history_check) = {yes: lecturer_notification}	detects that the student is disengaged, the student is assigned as motivational_break before returning to the loop. Finally, if the historical analysis proves a significant decrease or indicates a knowledge gap of the whole class, the algorithm achieves its objective by causing a lecturer_notification to be sent for the prompt teacher intervention.
Goal	lecturer_notification	The goal that we want to achieve is to notify the lecturer when a student shows disengagement during their learning process. This allows the lecturer to intervene and provide appropriate support to help the student get back on track with their studies.
Path cost	student_data_input, proficiency_score_evaluation, status_topic_weak, practice_module, engagement_check, assessment_history_check, lecturer_notification	The path cost is the execution time taken for the system to reach the goal which is to alert the lecturer about student disengagement based on engagement monitoring and assessment history checks.

1.3 Example solution:

1. student_data_input , proficiency_score_evaluation , status_topic_weak , remedial_module, engagement_check , assessment_history_check , lecturer_notification

1.0 PEAS Model Representation

Performance: How well students are learning, their engagement levels, improvement in scores, how quickly the system catches struggling students, accuracy in detecting weak areas, and how fast lecturers can respond to alerts

Environment: The Learning Management System (LMS), student databases, assessment records, attendance systems, classroom analytics, and past performance data

Actuators: Assigning remedial modules, delivering practice exercises, moving students to next topics, triggering motivational breaks, sending alerts to lecturers, and monitoring engagement

Sensors: Student data input, proficiency score checker, engagement detector (tracks attendance, participation, LMS activity), assessment history analyzer, and prerequisite checker

1.1 Explanation on each property in PEAS model

Performance Measure, Think of this as your system's report card. How do you know if it's doing a good job? You look at whether students are actually improving, staying engaged with their studies, and if lecturers are getting warned early enough when students start struggling. If the system can catch a student falling behind before it's too late, that's a win.

Environment is where the system "lives" and works. It's hanging out in the LMS, reading through grade databases, checking attendance records, and looking at how students have performed over time. The environment keeps changing because students' performance and engagement fluctuate throughout the semester.

Actuators are the system's hands and what it can actually do. When it spots a struggling student, it can assign them extra help through remedial modules. If they're doing okay, it gives them practice exercises. For the advanced students, it moves them to harder material. If someone's zoning out, it throws in a motivational break. And when things look serious, it rings the alarm bell to notify the lecturer.

Sensors are the system's eyes and ears and how it gathers information. It reads student data, analyzes test scores, watches engagement patterns, for instance, like who's showing up to class and participating, checks how students have been doing over time, and verifies if they have the right background knowledge. All this info helps the system decide what to do next.

2.0 Relations between each property

This section will show the relations between each property in the PEAS model including performance measure, environment, actuators, and sensors.

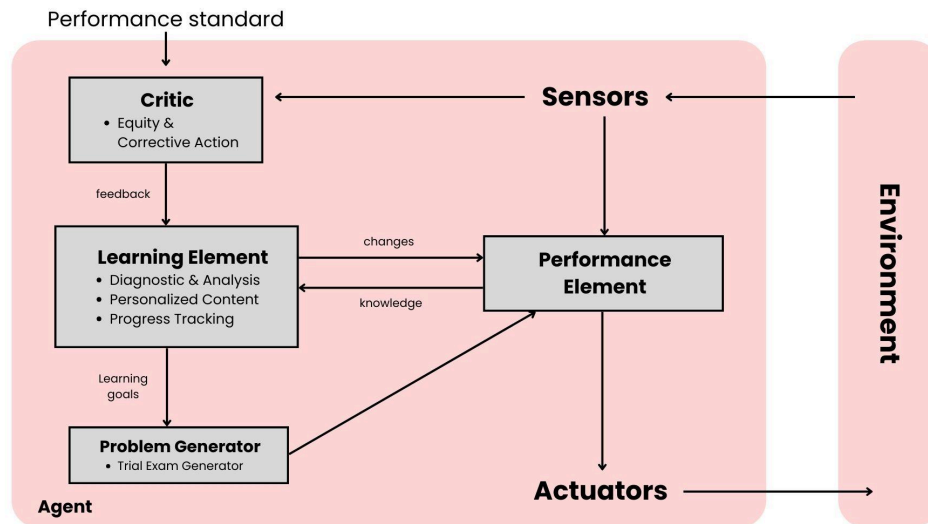


Figure 2.0 Intelligent Agent of Smart Education

The Figure 2.0 utilizes a Learning Agent Architecture which enables a continuous feedback loop between student data and system adaptation. The learning element serves as the system's brain by integrating the Diagnostic and Analysis, Personalized Content, and Progress Tracking agents. Sensors like marks of quizzes and student engagement levels will be analyzed to identify the knowledge gaps and provide the learning path that is suitable for the students. This helps to ensure the system meets its primary goals of early gap identification and maintain a flexible and personalized curriculum.

The performance element will execute the task such as suggest suitable modules and notifying the lecturers through the system's actuators. While the performance element handles the immediate execution of the student learning journey, the critic that is represented by the Equity and Corrective Action agent will help to monitor the long-term success. The critic will compare the student progress against the performance standard by detecting its pattern and send feedback to the learning element to trigger the corrective actions and additional support.

The problem generator consists of a Trial Exam Generator to challenge the student with the new assessments to test their "final exam readiness". The data that have been gathered from these trial examinations will feed back to the sensors which allows the system to refine its diagnostic accuracy.

In all, these components help to ensure the agent actively learns and improves to reduce performance variance across the student population.

3.0 Explanation in Proof of Concept (POC)

Proof of Concept (POC) demonstrate the feasibility of the suggested Smart Education Intelligent Agent by demonstrating how the system functions in a controlled academic environment utilizing actual student data inputs and simulated learning scenarios. The POC's goal is to confirm that the agent can successfully detect learning gaps early on, customize learning paths, let instructors track students' progress and lessen performance gaps.

Firstly, the Personalized Content Agent is triggered if knowledge gaps are identified. The bot suggests appropriate remedial modules, focused practice tasks, or advanced learning resources based on prerequisite mapping and sub-topic performance. Instead of a curriculum that is one-size-fits-all, this guarantees that every student obtains a personalized learning path. The main performance metric used to confirm the efficacy of this customisation is an improvement in post-module assessment scores.

Next, the Trial Exam Generator creates adaptive trial exams with past year questions and difficulty settings that are in line with students' areas of weakness in order to assess student preparedness and system correctness. The results are returned to the sensors for additional processing, and instant scoring enables prompt feedback. The dependability of the system's diagnostic skills is confirmed by looking at the relationship between trial exam performance and anticipated final exam preparation.

Furthermore, Progress Tracking Agent regularly monitors student involvement, including attendance, time spent on learning modules and participation in examinations,. Through the communication API, the agent notifies professors when abnormal patterns are identified, such as inactivity or persistent poor performance. This enables instructors to step in early, offer advice or provide more assistance before pupils drastically lag behind.

Last but not least, the Equity & Corrective Action Agent serves as the system's critic to guarantee equity and long-term progress. It finds growing differences between high-performing and struggling children by comparing student growth to predetermined performance standards. The agent autonomously modifies learning paths when performance variance is identified, giving weaker students more resources, more practice, or different explanations. The learning aspect receives feedback from this agent, enabling the system to continuously improve its tactics.

Overall, the POC verifies that the intelligent agent is capable of functioning as a closed-loop learning system. The system responds in real time to the needs of students through ongoing sensing, analysis, action, and feedback. The POC's findings show that the suggested design may accomplish the system's objectives of equitable academic achievements, early gap detection, individualized learning, and lecturer support.

4.0 Behaviours of Agent in AI to achieve The System Goals

Table 4.0 shows agents with their PEAS representation.

Agent	Performance measure	Environment	Actuators	Sensors
Diagnostic & Analysis Agent	Precision in identifying knowledge gaps. Accuracy of prediction to get an A	Student academic record. Grading schemes	Classifies topic status. Calculate target marks.	Quiz result. Test marks Historical grade data
Personalized content agent	Improvement in post-module scores	Library of sub-topic. Prerequisite mapping	Suggest specific modules	Prerequisite status. Sub-topic performance
Trial Exam Generator	Correlation between trial result and final exam readiness	Database of Past Year Question	Generate a Trial Exam. Provides instant scoring	Difficulty settings Students weak topic history
Progress Tracking Agent	Accuracy of student Effort reports. Reliability of readiness alerts	Lecturer dashboard. Communication API	Triggers lecturer notification Update students progress logs	Engagement checks Time spent on modules Final trial scores
Equity & Corrective Action Agent	Reduction in performance variance between students	Adaptive learning paths	Routes for struggling students to give an additional support	Repeated weak status inactive / struggle patterns

Table 4.0 agents with PEAS Model

Alignment of PEAS Model with the System Goals :

1. The Diagnostic agent uses its sensors to perform Intelligent Data Analysis. Its Actuator is a specific instruction on what the students are lacking. This aligns with goal 1 which is to Identify gaps early & calculate targets for students.
2. The personalized content agents act as the main job of the state space search where it ensures the learning journey is flexible based on students' needs and this agent is aligned with goal 2 which is to have personalized learning modules.

3. The progress tracking agent uses sensors to monitor students effort and only uses its actuator which gives notifications to the lecturers when students need help for their modules and this agent is aligned with goal 3 which is to assist lecturers in tracking progress.
4. The equity agent ensures that struggling students work harder by automatically providing them materials that they need in order to reach the advanced state. This agent will effectively close the knowledge gap where it aligns with goal 4 which is to reduce the gap between students who have difficulties in understanding and those who are advanced.